

Scenario 1: Manufacturing Quality Inspection

(a) Interpret the scenario and identify the type of noise present.

1. The manufacturing quality inspection system acquires digital images of metal components for automated defect analysis.
2. The presence of **repetitive horizontal stripe patterns** indicates **Periodic Noise**, which is commonly introduced by **electrical interference, electromagnetic disturbances, or power-line fluctuations** during image acquisition.
3. Periodic noise exhibits a regular and repeating pattern and is represented as distinct peaks in the frequency spectrum of the image

(b) Explain how this degradation affects inspection accuracy.

1. Periodic noise reduces the **signal-to-noise ratio (SNR)**, thereby degrading the overall image quality.
2. The horizontal stripe patterns may **obscure fine surface defects** such as micro-cracks, scratches, pits, and dents.
3. It introduces **false edge information**, leading to inaccurate edge detection and feature extraction.
4. Automated inspection algorithms may produce **false positives** (detecting defects where none exist) or **false negatives** (missing actual defects).
5. The degradation negatively impacts segmentation, pattern recognition, and classification accuracy, reducing the reliability of the quality inspection process

(c) Recommend an appropriate enhancement technique to remove the interference.

Recommended Technique: Frequency-Domain Notch Filtering

Processing Steps:

1. Convert the degraded image from the spatial domain to the frequency domain using the **Fast Fourier Transform (FFT)**.
2. Analyze the frequency spectrum to identify the high-energy frequency components corresponding to the periodic interference.

3. Apply a **Notch Reject Filter** (or multiple notch filters) to suppress the unwanted periodic frequency components while preserving the useful image information.
4. Perform the **Inverse Fast Fourier Transform (IFFT)** to reconstruct the enhanced image.
5. Normalize the output image to improve visual quality and prepare it for defect detection.

(d) Justify why your selected approach is more suitable than spatial-domain methods.

1. Periodic noise is localized at specific frequencies, making **frequency-domain filtering** highly effective for selective noise removal.
2. A **Notch Filter** attenuates only the unwanted frequency components while preserving important structural features such as edges, textures, and defect boundaries.
3. Spatial-domain filters such as **Mean Filter, Gaussian Filter, and Median Filter** primarily remove random noise and are ineffective against structured periodic interference.
4. Applying spatial filters to periodic noise often causes **image blurring**, loss of fine details, and degradation of edge sharpness.
5. Frequency-domain processing provides **higher restoration accuracy**, better preservation of image information, and improved performance for industrial inspection applications.

(e) Explain your answer in a logical sequence.

1. Acquire the metal component image using the automated inspection system.
2. Observe the presence of repetitive horizontal stripes and identify them as **Periodic Noise** caused by electrical interference.
3. Transform the image into the frequency domain using the **Fast Fourier Transform (FFT)**.
4. Detect the prominent frequency peaks associated with the periodic interference.
5. Design and apply a **Notch Reject Filter** to suppress the unwanted frequency components.

6. Reconstruct the filtered image using the **Inverse Fast Fourier Transform (IFFT)**.
7. Enhance the restored image through normalization or contrast adjustment, if required.

Scenario 2: Autonomous Vehicle Navigation

(a) Interpret the scenario and explain the challenge involved.

1. A self-driving (autonomous) vehicle relies on **computer vision systems** to capture real-time road images for lane detection, obstacle identification, and navigation.
2. During **heavy rainfall**, images become degraded due to rain streaks, water droplets, and sensor noise, which reduces overall image quality.
3. The primary challenge is to **suppress image noise while preserving critical structural features**, such as lane markings, road boundaries, traffic signs, and obstacle edges.
4. Excessive noise removal may blur these important features, leading to inaccurate perception and unsafe driving decisions.
5. Therefore, an effective enhancement technique must achieve an optimal balance between **noise reduction** and **edge preservation**.

(b) Identify the importance of preserving edges while removing noise.

1. Edges represent significant intensity discontinuities and define the boundaries of objects within an image.
2. Accurate edge information is essential for **lane detection, road segmentation, object recognition, and collision avoidance**.
3. Preserving edges ensures that lane markings remain clearly distinguishable even under adverse weather conditions.
4. Sharp edge information improves the accuracy of **feature extraction, image segmentation, and path-planning algorithms**.
5. Edge preservation minimizes false detections and enhances the reliability of the vehicle's perception system.
6. Maintaining structural details contributes directly to safe and efficient autonomous navigation.

(c) Recommend a suitable filtering technique.

Recommended Technique: Bilateral Filter (Edge-Preserving Smoothing Filter)

Processing Steps:

1. Capture the road image using the vehicle's onboard camera.
2. Apply the **Bilateral Filter** to the noisy image.
3. The filter computes pixel values by considering both:
 - **Spatial proximity (distance between neighboring pixels).**
 - **Intensity similarity (difference in pixel values).**
4. Smooth homogeneous regions while preventing smoothing across object boundaries.
5. Produce an enhanced image with reduced noise and preserved lane markings and road edges.

(d) Justify why your selected filter preserves important image details.

1. The **Bilateral Filter** performs **edge-preserving smoothing**, unlike conventional linear filters.
2. It combines **spatial weighting** and **intensity-based weighting**, ensuring that pixels with significantly different intensities are not averaged together.
3. As a result, important image features such as **lane boundaries, road edges, vehicles, pedestrians, and traffic signs** remain sharp.
4. Unlike the **Mean Filter** or **Gaussian Filter**, which often blur edge information, the Bilateral Filter effectively removes noise without degrading structural details.
5. It significantly improves the performance of downstream computer vision tasks, including **edge detection, lane tracking, object detection, and autonomous decision-making**.
6. The filter offers an excellent trade-off between **noise suppression** and **feature preservation**, making it suitable for real-time autonomous driving applications.

(e) Present your explanation clearly and systematically.

1. Capture the road image using the autonomous vehicle's vision system.
2. Observe degradation caused by heavy rain and sensor-induced image noise.
3. Identify the need to preserve lane markings and road boundaries during noise removal.
4. Apply the **Bilateral Filter** to perform edge-preserving smoothing.
5. Reduce noise while maintaining sharp object boundaries and structural features.
6. Generate an enhanced image suitable for lane detection, object recognition, and road segmentation.
7. Use the processed image for accurate navigation, obstacle avoidance, and safe driving decisions.